

Automated Attendance System Using Face Recognition

PHASE I REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Attendance management is an essential part of educational institutions for monitoring student participation and maintaining academic records. Traditional methods such as manual registers, RFID cards, and fingerprint-based systems can be time-consuming, require physical interaction, and depend on dedicated hardware. This paper presents an **Automated Attendance System using Face Recognition** that provides a contactless and efficient approach to attendance management. The proposed system uses a three-tier architecture consisting of a **React and Vite frontend, Node.js and Express backend, and Python Flask-based machine learning service**. The system supports separate roles for Admin, Teachers, and Students, providing features such as user management, class management, face registration, attendance tracking, leave management, and attendance reports.

For face recognition, **YuNet** is used for face detection and **SFace** is used to generate facial feature embeddings. During attendance, webcam frames are processed in real time, and detected faces are compared with registered student embeddings using **cosine similarity**. A similarity threshold of **0.48** is used to identify a matching student and automatically mark attendance. **MongoDB Atlas** is used to store users, classes, attendance records, and leave information, while **Cloudinary** is used for student photo storage. The proposed system aims to reduce

manual effort, minimize proxy attendance, eliminate the need for physical attendance hardware, and provide a centralized and scalable attendance management solution for educational institutions.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CORS	Cross-Origin Resource Sharing
FK	Foreign Key
HITL	Human-in-the-Loop
JSON	JavaScript Object Notation
JWT	JSON Web Token
ML	Machine Learning
PK	Primary Key
RBAC	Role-Based Access Control
REST	Representational State Transfer
RQ	Research Question
SPA	Single Page Application
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language

CHAPTER 1 INTRODUCTION

1.1 GENERAL

Attendance tracking is one of the simplest yet tedious tasks performed at educational institutions. Usually, students' attendance is tracked manually via attendance registers or sheets by calling their names and noting their presence in the class session. The traditional method, though widely used, has several shortcomings including wasting time during classes, possibility of proxy attendance wherein students mark attendance on behalf of absent ones, errors made during calculations, inability to maintain historical records, and the additional administrative load of faculty members.

With the development of computer vision and machine learning, biometric methods such as fingerprint scanning, iris scanning, and face recognition have been introduced as better options than the manual attendance tracking system. Face recognition system stands out among these as it is a simple process for the users since it requires neither physical interaction with the device nor queuing. The proposed system, an **Automated Attendance System using Face Recognition**, is a web/application-based solution designed to eliminate the drawbacks of manual attendance marking. The system is built around three distinct portals — **Admin**, **Staff (Teacher)**, and **Student** — each tailored to the specific role and responsibilities of its users.

- The Admin Portal will help administrators enroll teachers and students into the system, as well as associate students to their respective classes/subjects. This will help in keeping track of the core institutional data.

- The Staff Portal will allow teachers to see the students associated with their respective classes and then start the live face detection process during their lecture hours, where students coming past the camera will be automatically detected and marked as present. The portal will also give the flexibility to manually mark attendance whenever needed (such as camera failure, bad light, etc.).
- The Student Portal will allow students to enroll their facial information, which will later be used for detection. They will also be able to see the classes/subjects they are associated with as well as their attendance percentage on a subject by subject basis.

The use of face recognition technology together with the portal system will make the whole attendance procedure streamlined.

1.2 OBJECTIVE

The main goal of this project is to create and implement an automated, accurate, and efficient attendance system using the latest facial recognition technology. The specific goals of this attendance system include the following:

1. The creation of an attendance marking system which does not require any manual attendance and proxy attendance marking and which uses the facial recognition as the most reliable method of identity verification.
2. The development of an Admin portal for the enrollment of students and teachers in this system and linking students to specific classes/subjects.
3. The development of a Staff portal which allows teachers to see the list of students linked to particular classes, detect faces of students in real-time, mark attendance of students, and edit this attendance when it is required.

4. The development of a Student portal which will allow students to enroll their facial data, see all subjects where he/she is enrolled and see his/her attendance percentage in these subjects.
5. Saving time which is usually spent on attendance marking in order to increase the time of teaching.
6. Keeping all the attendance records safe and accurate.
7. To develop an attendance report which can help the institute analyze the trends in student attendance and take action on students with poor attendance.
8. To maintain secure data access, i.e., to limit access and manipulation of data to authorized personnel such as the admin, the concerned faculty or the student himself/herself.
9. To make the system scalable so that it can be adapted to other classes, departments or institutions with little change.

1.3 EXISTING SYSTEM

The system of attendance in educational institutions follows one of the below-mentioned systems:

- a) **Manual System** The teachers call the roll numbers or names of the students to record the attendance of the students in the register. This is the most popular attendance recording system even now.
- b) **RFID / Barcode-based Attendance Recording System** Students receive ID cards which have an RFID tag or barcode, which the students use to scan their attendance at the machine.

c) **Biometric Attendance Recording System** The fingerprint of the students is taken to mark their attendance in class.

Drawbacks of the Existing System

Timing issues: Both manual roll call method and swipe/scanning system take up a lot of time in the classroom.

Proxies for attendance: The manual and card system make it very easy for students to enter the attendance for a buddy who is not there.

Needs physical contact: The fingerprint and card system needs physical contact with the device that can create problems of hygiene and long waiting queues, especially when contagious disease is prevalent.

Errors: Human errors in recording and calculating the attendance percentage cannot be avoided.

Real-time information not available: Students do not have an easy access to know their current attendance percentage and have to approach faculty and office for this.

Difficult record maintenance: Physical registers can be lost, damaged, or difficult to search through, and consolidating records for reports is a tedious manual process.

No centralized system: Existing methods are often not integrated with a centralized platform where admins, teachers, and students can all access relevant information according to their roles.

Scalability issues: Hardware-dependent systems (RFID readers, fingerprint scanners) involve additional cost and maintenance and are difficult to scale across multiple classrooms.

These limitations highlight the need for a more efficient, contactless, and automated solution — which is addressed by the proposed **Face Recognition-Based Automated Attendance System** with its integrated Admin, Staff, and Student portals.

1.4 PROPOSED SYSTEM

The suggested system is a full-stack web-based Automated Attendance System with Facial Recognition running on a three-tier architecture which cleanly isolates the frontend, backend, database, and the facial recognition engine. The three portals (Admin, Staff, and Student) are accessed based on the role-based portal structure.

React.js has been used for creating the presentation layer, which includes the three role-wise portals – Admin, Staff, and Student – and interacts with the server via REST APIs. The application layer has been created using Node.js with Express.js framework that provides REST APIs for the purposes of authentication, enrollment, class mapping, marking attendance, and generating reports and also manages the flow of information among the frontend, database, and the facial recognition engine.

Central to the process is the face recognition pipeline, wherein OpenCV takes care of image pre-processing, U-Net based segmentation algorithm detects and localizes the face in the video feed, and the SFace model creates the face embeddings necessary for recognition. In the enrollment stage, a face of the student recorded through webcam is pre-processed and segmented using U-Net, followed by the

generation of the face embeddings through SFace, which are stored in MongoDB against the particular student's profile. In the live attendance phase triggered by a teacher, the same process repeats in order to extract embeddings from the video frame; however, the detected face embeddings are compared to those of students in the class using a threshold value and if a match is found, the backend makes an entry for the student in MongoDB along with other details such as student ID, class, date, time, and status without making any duplicate entry for the session. In case of failure of recognition because of low lighting condition or occlusion, or any other reason, the teacher can mark attendance manually through the Staff Portal.

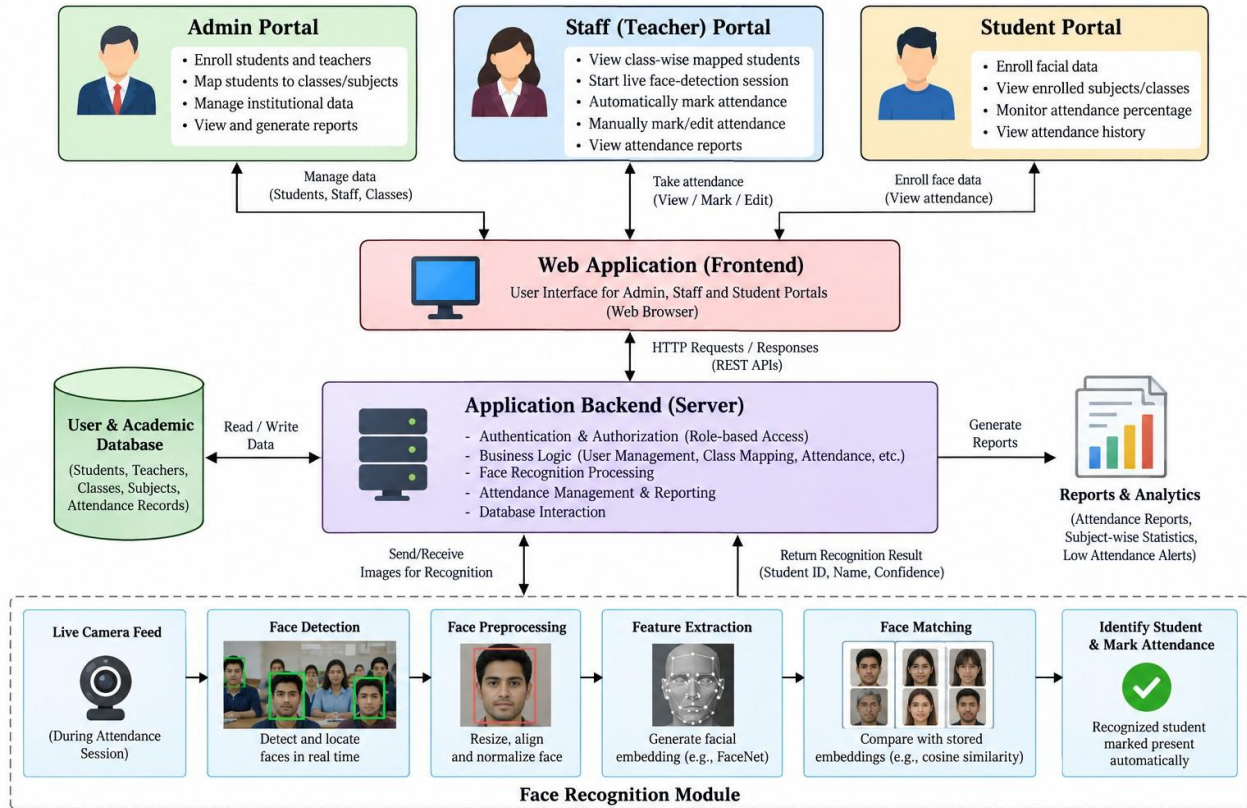


Figure 1.1: Proposed System Architecture of the Automated Attendance System using Face Recognition

Figure 1.1 shows the proposed architecture of the Automated Attendance System with Face Recognition. The proposed system comprises three portals namely, Admin, Staff, and Student, which are built using React.js and communicate with the backend Node.js/Express.js via REST APIs. Admin portal helps in managing the students, teachers, and classes whereas the Staff portal facilitates teachers to take attendance sessions, see students, and manage attendances manually. Students are able to enroll their facial data, view their subjects, and their attendance percentage through the student portal. All of these functionalities are linked to the central MongoDB database that stores user profiles, classes, facial embeddings, and attendance data.

The face recognition module is at the heart of the proposed system, as depicted in Figure 1.1. The process of enrollment includes the conversion of captured face image into a facial embedding using OpenCV, U-Net, and SFace. In live attendance, the same procedure is conducted on camera frames, and then a comparison is made between the generated embeddings and the student embeddings stored in the system through the use of a threshold. When there is a successful detection of a student, then the backend makes an entry for that particular student, class, date, and time of attendance. If recognition fails due to various reasons such as lighting, occlusions, and camera angle, the teacher can manually mark attendance via the Staff Portal.

CHAPTER 2 LITERATURE SURVEY

This chapter reviews published work in the four areas that WarrantyFlow draws on: LLM-based multi-agent orchestration frameworks, large-language-model extraction of structured data from invoices and similar business documents, machine-learning approaches to insurance and warranty fraud detection, and human-in-the-loop governance of agentic AI decision systems. Each strand is summarised on its own terms before Section 2.2 draws out what the combination means for a system like WarrantyFlow.

2.1 LITERATURE REVIEW

Traditionally, there have been many attendance systems, such as manual attendance systems, manual paper rolls, RFID cards, and fingerprint recognition systems, in educational institutes. But these methods are time-consuming, may contain human errors, and could be done by proxy attendance. Hence, face recognition has been developed as a non-contact biometric system for attendance automation and efficient attendance management [1].

Alhanaee et al. developed a Face Recognition Smart Attendance System based on Deep Transfer Learning, in which they had used pretrained Convolutional Neural Networks (CNNs) like SqueezeNet, GoogleNet, and AlexNet for the automatic process of attendance. In their research, it was shown that transfer learning can be utilized in reducing the number of training sessions but still has high recognition accuracy. They obtained the validation accuracies of 98.33%, 93.33%, and 100% for each of the three models, respectively.[2]

The Automatic Attendance System Using Face Recognition with Deep Learning Algorithm was developed by Al-Amoudi et al. by integrating MTCNN for face detection and FaceNet for face recognition. The system uses 908 images of 21 participants in which 108 images were used for the testing process. The accuracy achieved in this regard was 100% in case of face detection and 87.03% in case of face recognition. The success of this approach proves the viability of using face detection and face recognition based on deep learning for attendance [3].

The Development of Automated Attendance System Using Pretrained Deep Learning Models was studied by Ahmad et al. The study tested six different pretrained face recognition models and chose FaceNet for the automatic attendance system. In this regard, the accuracy achieved by the selected model was reported to be more than 95% [4].

Amri et al. suggested a Real-Time Face Attendance System Using CNN MobileNet and MTCNN, which comprised CLAHE image preprocessing, MTCNN face detection, and MobileNet face recognition. The research was dedicated to real-time attendance with taking into account variations in illumination, face alignment, and performance. This project presented an example of practical application of image preprocessing, face detection, and recognition algorithms for classroom attendance [5].

Singh and Sharma created a Real-Time Face Recognition for Automated Attendance Management system that comprised MTCNN, FaceNet, and ArcFace algorithms. The dataset used for the experiments consisted of 4,200 pictures of 120 students. The research achieved 98.7% of recognition rate with possibility to recognize simultaneously more than one face. This example shows the significance of multi-face recognition for real-time classroom environment [6].

A system named Multi-Facet Facial Recognition Model for Institutional Attendance Activities During Examination Scenario was designed by Jack et al. It uses Haar Cascade for face detection, ResNet based 128 dimensional feature extraction, and KNN classifier. The model performs evaluation using static images, videos, and live stream cameras and has been reported to achieve 99% accuracy while taking into account the effects of lighting, orientation of head, presence of glasses and beards [7].

Budiman et al. performed a systematic review of literature comparing performance of LBPH and CNN based methods in student attendance application. As per their review findings, CNN based models usually offer better and consistent recognition accuracy compared to LBPH, especially when affected by external conditions on image quality. The study reflects the evolution from conventional methods to deep learning based face recognition systems for attendance applications [8].

The U-Net architecture was introduced by Ronneberger et al. for image segmentation tasks. This architecture uses an encoder-decoder design with skip connections to provide accurate pixel level segmentation. Even though designed for biomedical image segmentation, the architecture can serve as a base for future image segmentation based computer vision applications [9].

Zhong et al. developed SFace (Sigmoid-Constrained Hypersphere Loss) for robust face recognition. The model considers the impact of poor quality and noise in the training set by controlling intra-class and inter-class optimization. The authors assessed SFace on several face-recognition benchmark datasets, showing robustness improvements. Thus, SFace is highly applicable to the proposed system that generates discriminative face representations under realistic conditions [10].

The current literature clearly illustrates a progressive transition from classical face-recognition approaches like Haar Cascade and LBPH to modern deep learning approaches like MTCNN, FaceNet, ArcFace, ResNet, and SFace. The existing research shows promising face recognition capabilities; however, there are still problems in the real-world classroom environment due to illumination variation, occlusion, head pose, simultaneous multi-faces recognition, computational demands, and privacy concerns.

The above discussion justifies the need for designing the proposed Automated Attendance System Using Face Recognition which includes the use of OpenCV, U-Net, and SFace in the complete role-based web application. While the existing systems mainly concern face detection and recognition, the proposed system incorporates student registration, class assignment, live face recognition, automated attendance recording, manual correction of attendance and attendance report using Admin, Staff, and Student portals through React.js, Node.js/Express.js, and MongoDB.

2.2 COMPARISON AND DISCUSSION

There is clearly observed evolution in the field of attendance control systems, which use faces as an identifier, from conventional computer vision algorithms to deep learning-based detection and recognition approaches. The former such as Haar Cascade and LBPH do not demand a lot of computing resources, but they are sensitive to the variations of lighting, head position and face occlusion. The latter including MTCNN, CNN, FaceNet, ArcFace, and ResNet ensure better representation of facial features and recognition quality, though they usually demand more computing resources [2]–[8].

Moreover, the studied research papers differ in terms of the number of samples in the dataset and the presence/absence of real-time face recognition. Al-Amoudi et al. used the small-size sample of 21 people, whereas Sharma and Singh used 4,200 images of 120 people and performed simultaneous multi-face recognition [3], [6]. Similar considerations have been taken into account in several other studies. It should be noted that the factors mentioned above influence on the system performance.

Study	Detection / Segmentation	Recognition / Classification	Dataset / Evaluation	Main Focus
Alhanaee et al. [2]	CNN-based	SqueezeNet, GoogleNet, AlexNet	Student attendance	Transfer learning
Al-Amoudi et al. [3]	MTCNN	FaceNet	21 students	Automated attendance
Ahmad et al. [4]	Face detection	Pretrained deep-learning models	Attendance dataset	Accuracy and computational efficiency
Amri et al. [5]	MTCNN + CLAHE	MobileNet	Real-time attendance	Efficiency and illumination
Sharma and Singh [6]	MTCNN	FaceNet + ArcFace	4,200 images / 120 students	Multi-face recognition
Jack et al. [7]	Haar Cascade	ResNet + KNN	Static images, video, and live feeds	Environmental variation
Budiman et al. [8]	LBPH / CNN	LBPH / CNN	Systematic literature review	Comparison of approaches
Proposed System	U-Net	SFace	Real-time classroom environment	Integrated automated attendance

Table 2.1: Comparative Analysis of Attendance Systems

One of the gaps noted from the comparison of the research work is that the majority of studies focus mainly on the face detection and recognition algorithm itself. Other vital elements like student enrollment, class mapping, role-based access management, live attendance, duplicate prevention, manual attendance update, and real-time attendance reports are not covered in a unified system.

The proposed system solves this gap with the integration of U-Net-based face segmentation with SFace-based facial embedding along with integrating the whole recognition pipeline within a complete web application. For this purpose, React.js will be used as the frontend framework, Node.js/Express.js will be used as the backend, and MongoDB will be used for the central database. It also includes three different portals – Admin, Staff, and Student, allowing the system to perform the entire process from facial-data enrollment to real-time recognition and attendance updates.

2.2.1 RESEARCH GAP

Despite many researchers developing face recognition based attendance systems, there are still some challenges that exist. Existing works usually focus more on face detection and face recognition accuracy and do not pay enough attention to the actual application of the system in the classroom setting with several students. There are also problems related to the use of small data sets for testing and evaluation of the algorithm.

Also, in existing works, there is a lack of the implementation of the entire process of attendance. Most works do not incorporate the processes of student registration, class/subject mapping, role-based authorization, real-time attendance, duplication avoidance, manual correction and reporting in one system. Computational efficiency on standard hardware is another requirement.

Hence, the proposed work will address the gaps in the literature by integrating U-Net and SFace algorithms and the web-based platform with Admin, Staff and Student portals into one system.

2.2.2 PROPOSED CONTRIBUTION/NOVELTY

This system presents a novel real-time attendance system using facial recognition that integrates face recognition with the role-based web application. The system presents following novelties:

U-Net and SFace integration: U-Net is used for face detection/segmentation, whereas SFace is responsible for generating the facial embedding.

Real-time attendance: Automatically identifies the presence of the students through live camera frames and logs their attendance.

Role-based access control: Three different portals for Admin, Staff, and Student for users, classes management, attendance, and records management respectively.

Class and subjects mapping: Mapping of students and staff for the corresponding class/subjects prior to attendance process.

No duplication: Same student cannot be marked multiple times during the attendance session.

Manual fallback: In case of failure due to lighting, occlusion, or camera angle, the staff member can manually mark the attendance.

Centralized attendance management system: Storage of student information, class information, facial embeddings and attendance data into MongoDB database.

In conclusion, novelty is based on the integration of the face recognition pipeline along with the complete attendance management system via web-based role-based approach.

2.3 CONCLUSION

The implementation of the Automatic Attendance System through face recognition offers a highly efficient and automatic way of dealing with the inefficiencies associated with traditional attendance mechanisms. The proposed attendance system will use U-Net for face detection/segmentation and SFace for face recognition that will help identify the students through live video capture and attendances recorded.

The use of the Admin, Staff, and Student portals will enable the system to have a central mechanism for handling the registration of students, mapping students to classes and subjects, recording of the attendance, and generating reports. The system also ensures that there is a way of preventing duplication and correcting manual attendance in case of environmental constraints like poor lighting.

In summary, the proposed system will offer an efficient and scalable way of ensuring attendance in classrooms.

CHAPTER 3 SYSTEM DESIGN

This chapter describes the complete architecture, software and hardware requirements, database design and deployment architecture of the proposed Smart Attendance System. The system is implemented as a full-stack web application with a React-based user interface, a Node.js and Express backend, a MongoDB database and an independent Python-based facial recognition microservice . The architecture separates application management, data persistence and computer-vision processing into independent layers to improve modularity, maintainability and scalability.

3.1 SYSTEM ARCHITECTURE

The proposed Smart Attendance System adheres to a dedicated machine learning service in a three-layer full-stack architecture . The main layers of the system are Frontend Layer, Backend/Application Layer, Database Layer and Machine Learning Layer .

The frontend is built with React and has distinct interfaces for Administrator, Teacher and Student roles. The backend is developed using Node.js and Express.js and serves as the main application server for authentication, user and class management, attendance management, leave and on-duty requests, and communication with the machine learning service. MongoDB (using Mongoose) is used by the application server for persistent data storage.

The machine learning layer is implemented as a standalone microservice on Flask. It gets images or video frames from the backend, and detects and recognizes faces with OpenCV YuNet

Authentication and Role Management

The authentication subsystem uses **JWT-based authentication**. During login, the backend verifies the user's credentials and generates a signed JWT containing the user's ID, role, and department. The token is subsequently used to access protected resources.

The system supports three user roles:

- **Administrator**
- **Teacher**
- **Student**

The frontend implements role-protected routes so that users are directed only to the pages permitted for their respective roles. The React routing configuration contains separate route groups for /admin, /teacher, and /student.

Face Recognition Architecture

The face recognition pipeline is implemented as a standalone Python Flask microservice. The service takes a frame, decodes it into an OpenCV image, finds faces, extracts face embeddings, and compares them against the enrolled class database.

The recognition process consists of the following stages:

Image Capture → Frame Transmission → Face Detection → Image Enhancement → Face Alignment → Feature Extraction → Embedding Comparison → Student Identification → Attendance Recording

The system detects faces by using YuNet and extracts facial features by using SFace/ArcFace . The recognition engine also includes CLAHE-based illumination

normalization, image sharpness check, multi-vector embeddings, centroid representation, and multi-face recognition.

Setting the face matching similarity threshold to 0.48. The system can recognize multiple faces in the same frame, and give the detected student's roll number, name, department, bounding box, and confidence information.

Class-Based Recognition

Each time the class attendance session is taken, the system stores a temporary facial database separately. When a teacher starts a session, the backend gets the students for the selected class and syncs their stored embeddings with the ML service.

In this way, the system will only recognize the students of the selected class, and will not compare the captured face with all the students of the institution. It helps to eliminate unnecessary comparisons, and makes sure attendance is attributed to the right class.

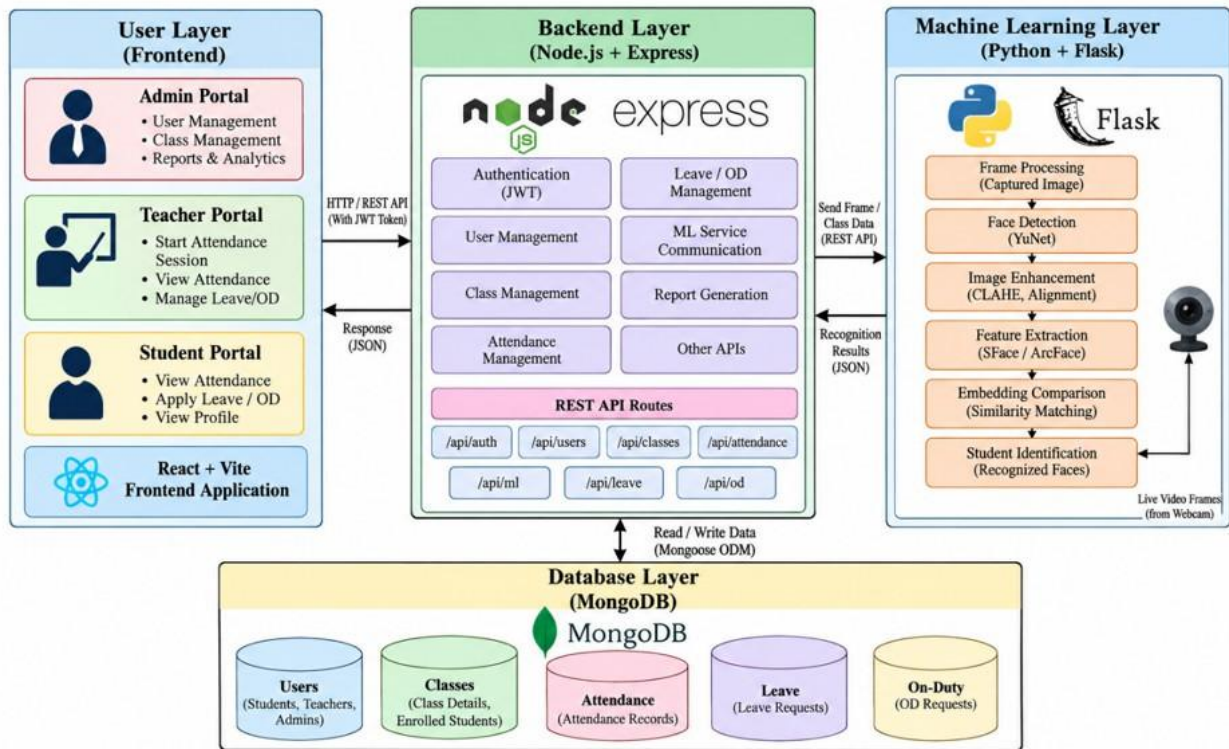


Figure 3.1. System architecture diagram of the proposed Smart Attendance System

Figure 3.1. System architecture diagram

3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

The Smart Attendance System is implemented using a combination of JavaScript-based web technologies and Python-based computer vision technologies.

Backend Software Stack

The backend is developed using **Node.js** and **Express.js**. Express provides the REST API layer through which the frontend communicates with the backend. MongoDB is used as the primary database, with Mongoose providing object-document modelling and database interaction.

Component	Role
Express.js	REST API and application server
MongoDB	Persistent database for users, classes and attendance
Mongoose	Object Data Modeling layer for MongoDB
JWT	Authentication and authorization
bcryptjs	Password hashing and verification
Axios	Communication between backend and ML service
Multer	Handling uploaded files/images
Cloudinary	Cloud-based image storage
ExcelJS	Excel-based attendance/report processing
CORS	Cross-origin communication between frontend and backend
dotenv	Environment variable configuration
Express.js	REST API and application server
MongoDB	Persistent database for users, classes and attendance
Mongoose	Object Data Modeling layer for MongoDB
JWT	Authentication and authorization

Table 3.1. Backend software stack and libraries

Frontend Software Stack

The frontend is implemented as a React-based single-page web application using Vite as the development and build environment.

Component	Role
React	Component-based frontend development
React DOM	Rendering React components
Vite	Development server and production build tool
React Router DOM	Client-side routing
Axios	Communication with backend APIs
Tailwind CSS	User interface styling
Recharts	Attendance analytics and graphical reports
Lucide React	UI icons
React	Component-based frontend development
React DOM	Rendering React components
Vite	Development server and production build tool
React Router DOM	Client-side routing
Axios	Communication with backend APIs

Table 3.2. Frontend software stack and libraries

Machine Learning Software Stack

The face recognition component is implemented as a Python Flask microservice.

Component	Role
Python	Machine learning and image-processing implementation
Flask	ML REST API service
OpenCV	Image processing and face recognition
YuNet	Face detection
SFace / ArcFace	Face feature extraction and recognition
NumPy	Numerical operations and embedding calculations
ONNX Models	Execution of face detection and recognition models
Pickle	Temporary class-specific embedding database
python-dotenv	Environment configuration
Python	Machine learning and image-processing implementation
Flask	ML REST API service
OpenCV	Image processing and face recognition
YuNet	Face detection

Table 3.3. ML stack and libraries

3.2.2 HARDWARE REQUIREMENTS

The system requires a computer system that can run React frontend, Node.js backend, MongoDB database, and Python based machine learning service. Attendance session needs a real time face capture device, such as a webcam.

Component	Recommendation
Processor	Intel Core i5 / AMD Ryzen 5 or above
RAM	Minimum 8 GB; 16 GB recommended
Storage	Minimum 10 GB free storage
Camera	HD webcam, preferably 720p or above
Network	Stable LAN/Wi-Fi connection
GPU	Not mandatory; optional for future acceleration

Table 3.4. Hardware / deployment requirements

3.3 DATABASE DESIGN

The Smart Attendance System uses MongoDB as its main persistent database. The backend is made with Node.js and communicates with MongoDB using Mongoose. The database stores user's information, class information, attendance records and other academic workflow information.

The main entities of the system are User, Class, Attendance, Leave and On-Duty records. Part of the user information also stores the facial embeddings associated with students and is synchronized with the ML service when needed. User model

contains role information, student roll number or teacher employee id, department, profile photo information and facial embeddings.

Entity	Purpose
User	Stores administrator, teacher and student information
Class	Stores class, subject, department, teacher and enrolled student information
Attendance	Stores attendance status, date, timestamp, confidence and verification method
Leave	Stores student/teacher leave requests and related information
On-Duty	Stores on-duty requests and approval information
Facial Embeddings	Stores numerical facial representations associated with students

Table 3.5. Database entities and their purpose

3.4 DEPLOYMENT ARCHITECTURE

- The Smart Attendance System is implemented as a set of loosely coupled services that communicate using REST APIs.
- The React frontend is the application that the user interacts with. It communicates with the Node.js backend over HTTP requests. The Node.js/Express backend is the primary application server, and interacts with MongoDB for persistent storage. It also serves as a proxy between the frontend and the Python ML service.

- The Python Flask ML service is a standalone service that exposes APIs for face enrollment, class DB sync, session management and real-time face recognition. By default, the backend configuration points to the ML service at <http://127.0.0.1:5001>.

3.5 ACTIVITY DIAGRAM

Activity diagram is a representation of the sequence of operations performed in the Smart Attendance System. The process starts when the user enters login credentials, then the system verifies the credentials and determines the role of the user as Administrator, Teacher or Student. Based on the role, the user is redirected to the respective dashboard and the allowed operations are provided.

The teacher picks a class and initiates an attendance session for automatic attendance. The system loads the enrolled students and their face embeddings. The classroom camera captures live frames. The machine learning service uses YuNet for face detection and SFace/ArcFace for feature extraction and recognition to analyze the frames. The identified student is mapped with the enrolled data and the attendance record is stored in MongoDB with date, timestamp, confidence and verification method. Updated attendance information is reflected on the teacher and student dashboards.

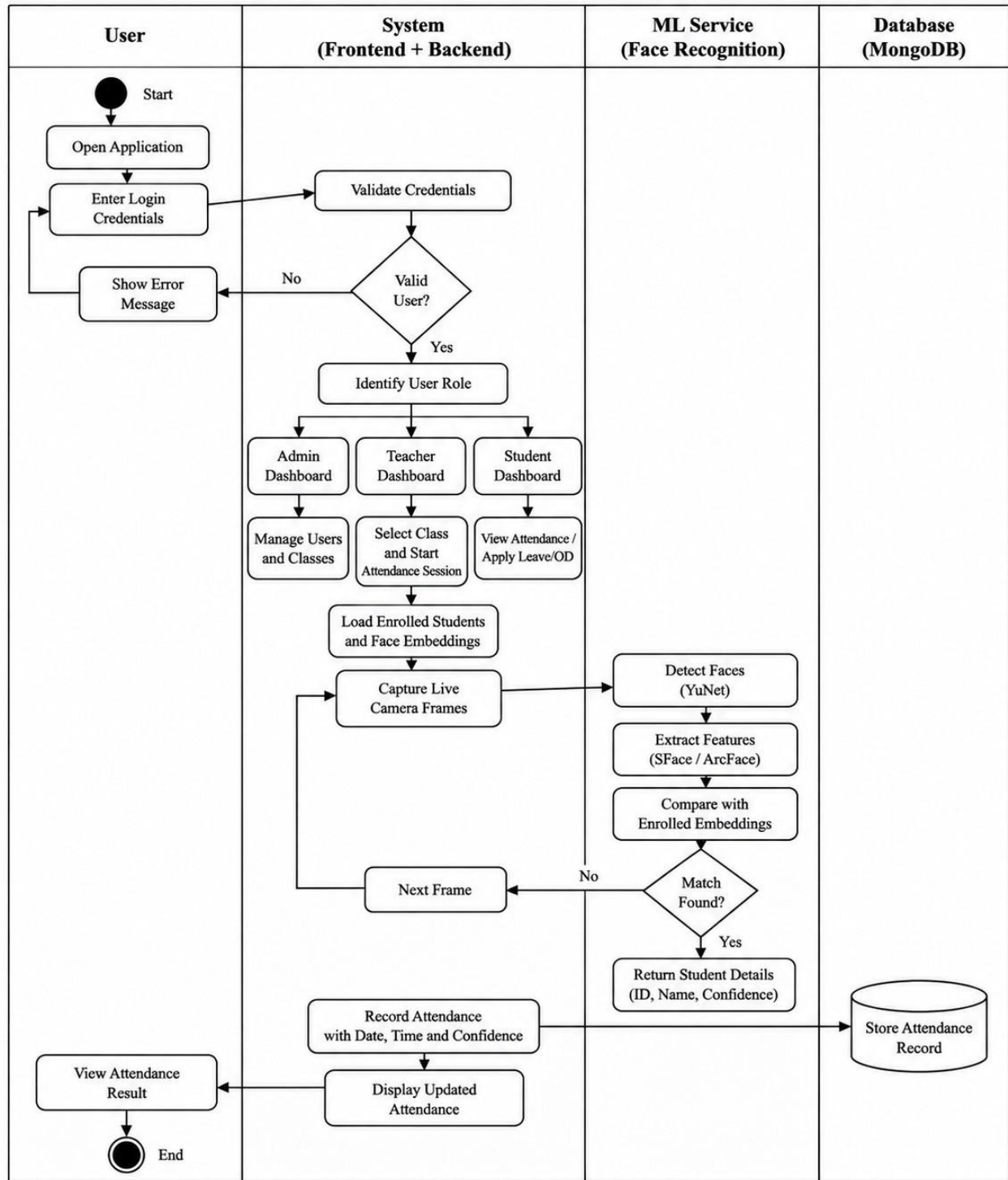


Figure 3.2. Activity diagram of the Smart Attendance System

3.6 DATA FLOW DIAGRAM

The Data Flow Diagram illustrates the flow of information between users, application processes, the machine learning service and the database.

Administrator: User and class management Teacher: Attendance session management and view attendance records The system allows students to view their attendance and apply for leave or on-duty.

For face-based attendance, the backend feeds live camera frames to the machine learning service. The ML service then detects faces, extracts facial features and compares the features with the enrolled student embeddings. Then the backend receives the recognition result and saves the attendance to MongoDB. Then the stored information is retrieved and presented via the appropriate dashboards and reports.

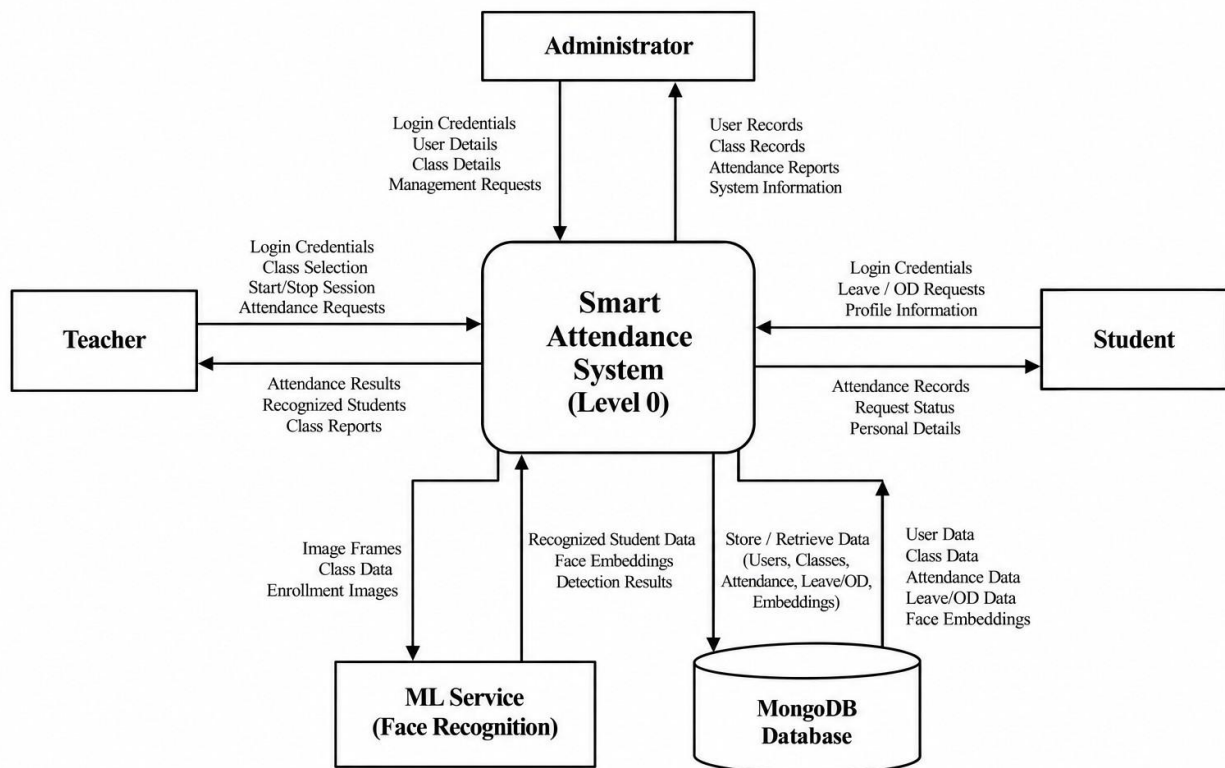


Figure 3.3 Data flow diagram of the Smart Attendance System

CHAPTER 4 PROJECT DESCRIPTION

4.1 METHODOLOGY

Smart Attendance System using Face Recognition is a full-stack application designed to make the process of marking student attendance automatic and easier. This application incorporates a frontend application built using ReactJS, a backend using NodeJS and ExpressJS, a database using MongoDB, and a Python based machine learning service for managing users' data, enrolling faces, recognizing faces, and attendance records. The entire process starts with registering the user and enrolling their faces, after which their facial data is stored for future recognition. In a live attendance session, the teacher chooses the necessary class and turns on the camera, which recognizes and identifies the enrolled students through the detected faces. If the recognition is done successfully, attendance is registered without any duplicate entry within the same session. The attendance data will be centrally stored and retrieved, modified, analyzed, and exported as needed. There are three portals in this application depending on the roles, which include Admin, Teacher, and Student.

4.2 MODULES

4.2.1 FACE ENROLLMENT AND RECOGNITION

The Face Enrollment and Recognition Module constitutes the backbone of the proposed attendance system. Upon the enrollment process, the system extracts the facial image of the student and analyzes it to produce a unique facial signature. For face detection and facial embedding extraction, YuNet and SFace services are used

respectively. Facial image preprocessing through illumination normalization and image quality verification is performed to enhance the quality of the facial data. For the attendance process, the detected face is transformed into a facial embedding and is matched against the facial signatures of the enrolled students within the selected class. Cosine similarity is applied for the matching process along with a predetermined similarity threshold to identify whether the detected face belongs to one of the enrolled students. Multiple faces can be processed per frame.

4.2.2 LIVE ATTENDANCE AND SESSION

The Face Enrollment and Recognition Module constitutes the backbone of the proposed attendance system. Upon the enrollment process, the system extracts the facial image of the student and analyzes it to produce a unique facial signature. For face detection and facial embedding extraction, YuNet and SFace services are used respectively. Facial image preprocessing through illumination normalization and image quality verification is performed to enhance the quality of the facial data. For the attendance process, the detected face is transformed into a facial embedding and is matched against the facial signatures of the enrolled students within the selected class. Cosine similarity is applied for the matching process along with a predetermined similarity threshold to identify whether the detected face belongs to one of the enrolled students. Multiple faces can be processed per frame.

4.2.3 ATTENDANCE MANAGEMENT

The Attendance Management Module controls the process of creating, storing, retrieving, and updating attendance logs. Each log holds pertinent data about the student, his/her class, status of attendance, the date and time, and other details related to the process including the confidence level of the scan and the verification method. Moreover, there is a function of manual override for the teachers when

automatic recognition fails because of factors like insufficient lightening or incorrect face position. There is also an option to export attendance logs in CSV and Excel format.

4.2.4 AUTHENTICATION AND ACCESS CONTROL

The Authentication and Access Control module ensures secured access to the system. The user is verified through the login process and a JSON Web Token is generated if successful in authentication. This token is used to authenticate the restricted resources and identify the user role while using the application. The application has three key roles including Administrator, Teacher, and Student. The Administrator has privileges to manage the users and classes; teachers have privileges to conduct and manage attendance; and the students have privilege to access their attendance data only. Role-based access control ensures the users access the application features according to their role only.

4.2.5 ATTENDANCE ANALYTICS AND REPORTING

The Attendance Analytics and Reporting Module turns the stored data into meaningful attendance statistics for the use of teachers and administrators. The module supplies attendance statistics and course summaries to help the users analyze the attendance trends and track the attendance history of their students. Furthermore, the module is capable of generating reports in the CSV and Excel format. Such reports can serve as an aid for documenting the attendance and maintaining the records. Through the attendance statistics and report generation module, the manual preparation of the attendance data becomes unnecessary.

4.2.6 ROLE-BASED FRONTEND PORTALS

Frontend of Smart Attendance System is designed as a single react based application offering a different user interface for each of the user role – admin, teacher, and student. Every role has been assigned a distinct dashboard with navigation limited only to those functionalities that are relevant for that specific user. The front end communicates with back end using REST APIs to fetch and update the data related to users, classes, attendance, and face recognition.

Admin Layout (also known as Admin portal) is considered to be the administrative control center in the system which gives an insight into the entire attendance system infrastructure. The details that are visible in the dashboard include status of the Node REST API, MongoDB connectivity, face AI engine, user registration, active classes, attendance details, and attendance percentage, as illustrated in Figure 4.1. Further, the administrator can visit the User Directory and Biometrics module to manage the students and faculty that have enrolled in the system. The details provided here include student's department, roll number, biometric vector availability, and account status, among other details, as shown in Figure 4.2.

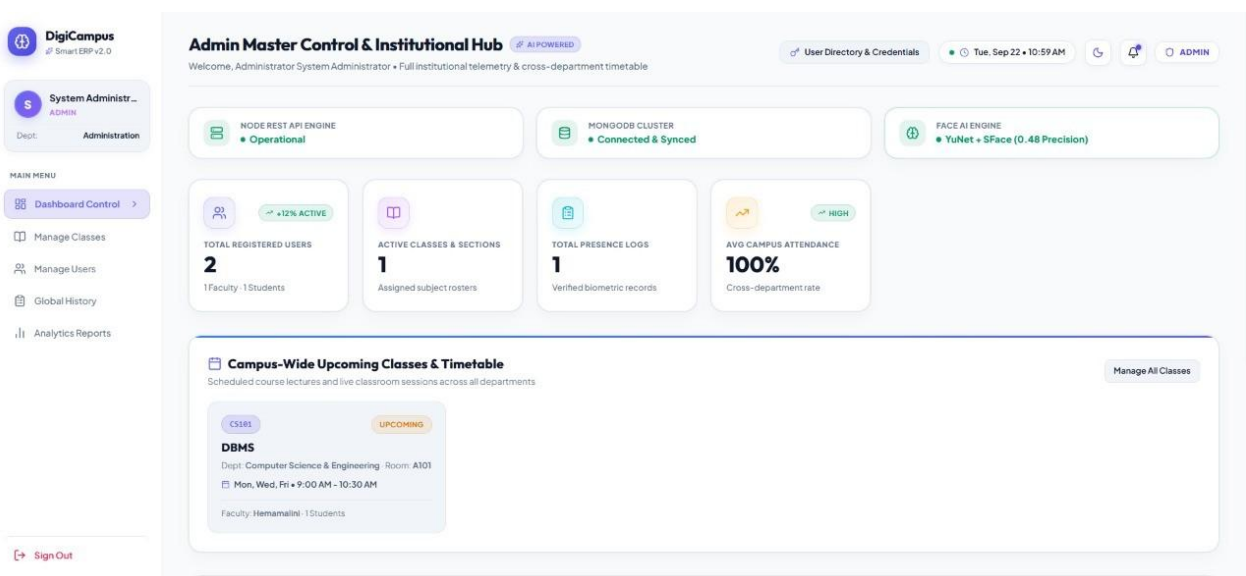


Figure 4.1. Admin Master Control Dashboard

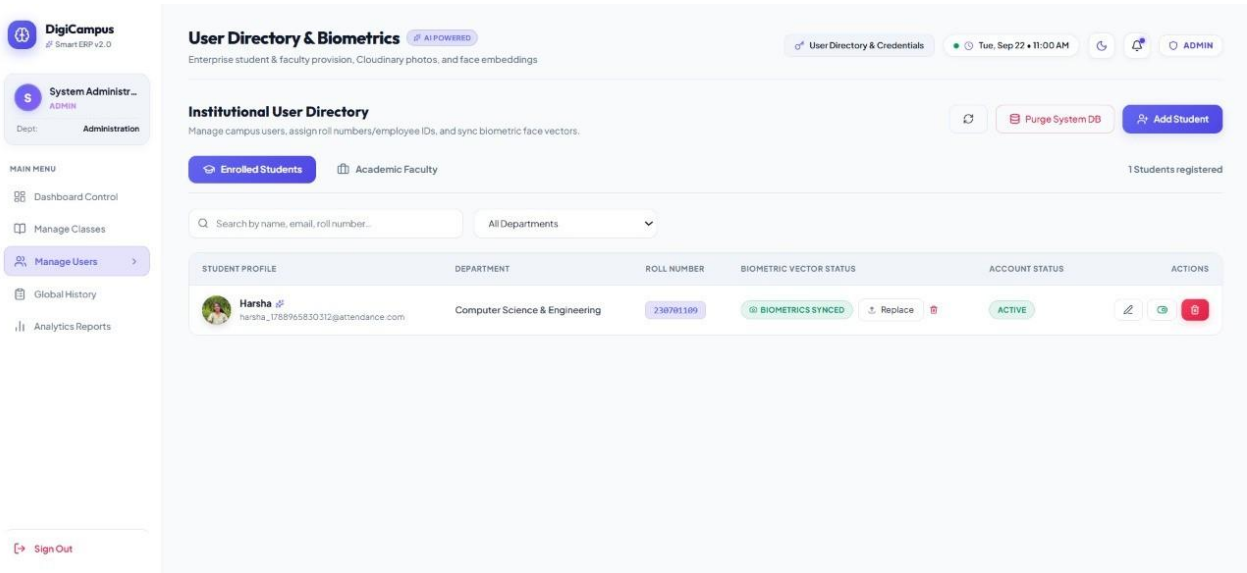


Figure 4.2. User Directory and Biometrics

Through the Student Portal (StudentLayout), students get an individualized dashboard that enables them to track their attendance and schedule. The dashboard shows the attendance percentage of the student, exam eligibility status, total number of sessions, number of present sessions and absent sessions, and attendance percentage of individual courses, as indicated in Figure 4.3. Students are also able to navigate into the Attendance History feature, which gives them details about the classes yet to come, courses that the students are registered for, their timetable per day, and the attendance details on a per-course basis, as seen in Figure 4.4..

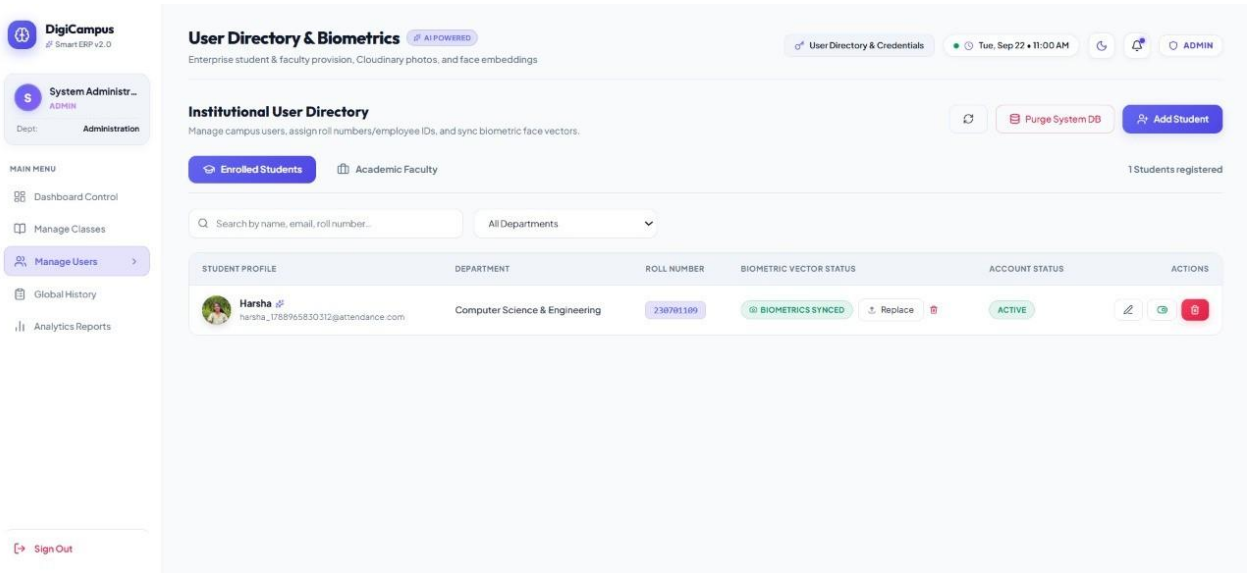


Figure 4.3. Student Dashboard and Attendance Portal

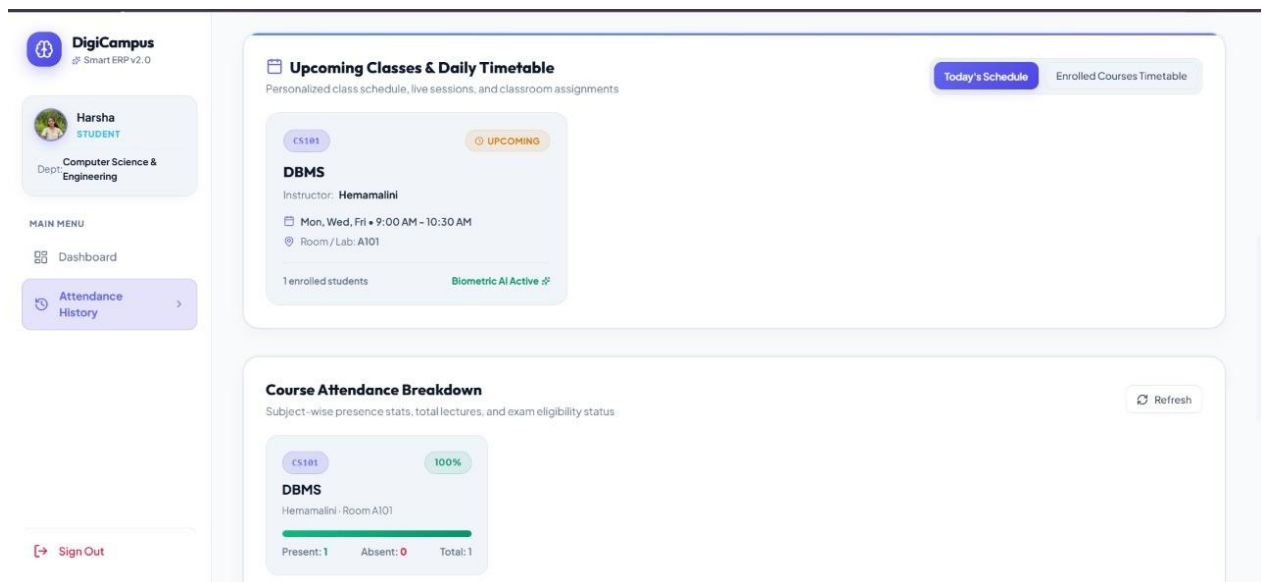


Figure 4.4. Student Attendance History

TeacherLayout is the Teacher portal that enables all necessary actions in order to manage the attendance of classes. The Teacher portal consists of accessing the

following: teacher dashboard, Live AI Camera & Hybrid Session Console, class roster, and course reports. The Live AI Camera & Hybrid Session Console allows a teacher to choose a section of a course and create a live attendance session via the face recognition of the students. The recognized students are displayed in the AI Frame Detection Log, where the attendance can be checked prior to sending the report (refer to Figure 4.5). The Teacher portal also contains the Analytics & Export Reports option, where teachers are able to see general information about the attendance of the courses, including events, sections, trends, and statistics (present/absent). Attendance can also be exported in CSV and Excel formats (refer to Figure 4.6)..

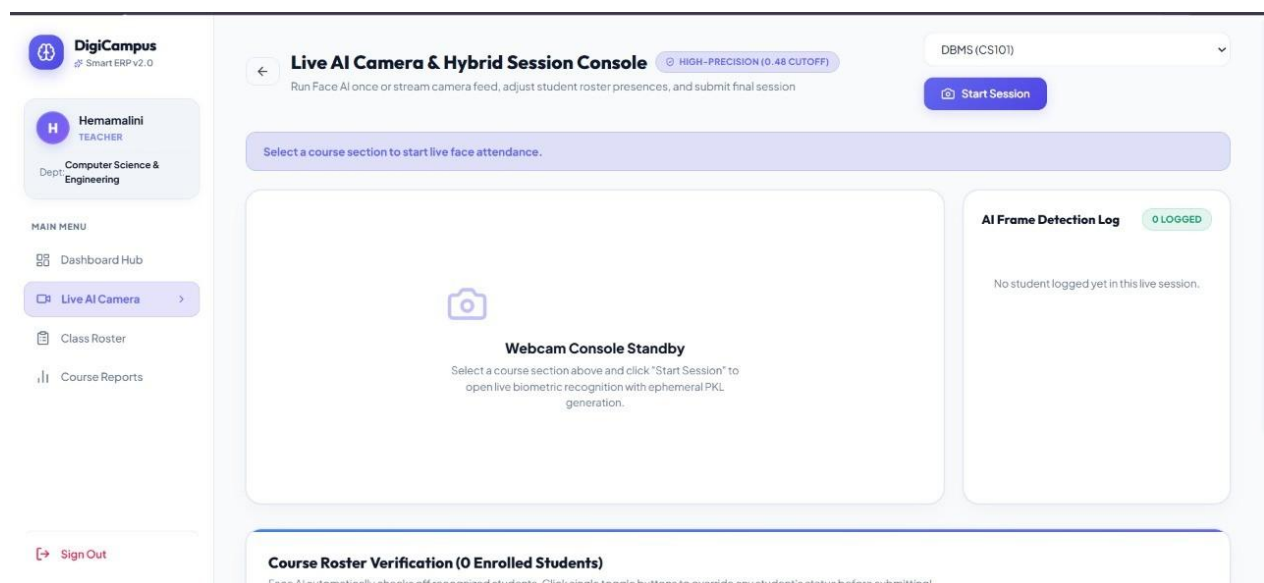


Figure 4.5. Teacher Live AI Attendance Session

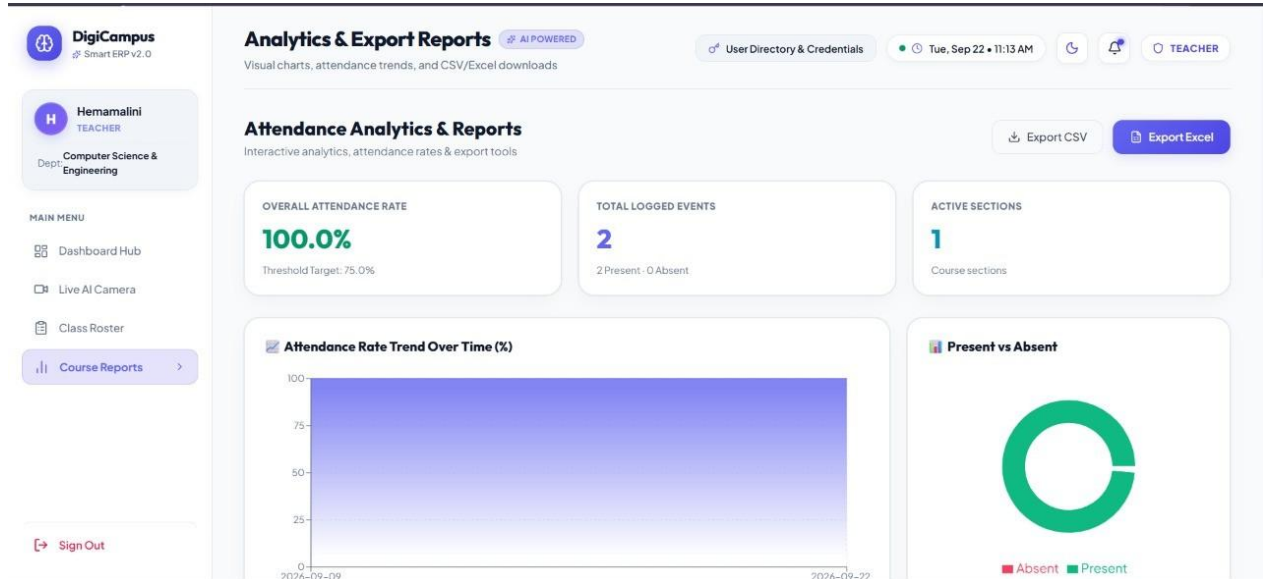


Figure 4.6. Attendance Analytics and Export Reports

These role-based portals are integrated with the common backend services as well as centralized database and therefore the information which is produced during the process of student enrollment, biometrics, face recognition, and attendance management sessions would be displayed through the corresponding interfaces.

The admin portal is responsible for system and user management while the teacher portal is designed for classroom attendance management. The student portal, on the other hand, is responsible for student attendance management.

CHAPTER 5 IMPLEMENTATION AND RESULT DISCUSSION

This chapter documents the executed REST API endpoints and presents the working Admin portal as evidence of the system's end-to-end capability. It describes the primary modules of the attendance management system, comprising student and instructor management, face registration, face detection and recognition, attendance marking, and centralized attendance record management. The chapter moreover discusses the experimental evaluation of the proposed system, comprising face detection accuracy, face recognition performance, attendance marking accuracy, processing time, baseline comparison, error analysis, and the limitations of the evaluation.

5.1 API ENDPOINT SUMMARY

The backend exposes REST API endpoints that support the essential procedures of the Smart Attendance Management System. These endpoints handle user authentication, student and educator management, face registration, face recognition, attendance marking, and retrieval of attendance records. The APIs are integrated with the React frontend and the Python-based face recognition module to provide a total and whole and exhaustive and inclusive end-to-end attendance management workflow

Method & Path	Description
---------------	-------------

POST /api/auth/login	Authenticates a user and provides the authentication token.
GET /api/auth/me	Retrieves the details of the currently authenticated user.

Table 5.1. Authentication API endpoints

Method & Path	Description
GET /api/users	Retrieves the user directory for Admin and Teacher users.
POST /api/users	Creates a new user.
GET /api/users/:id	Retrieves details of a specific user.
PUT /api/users/:id	Updates the details of an existing user.
DELETE /api/users/:id	Deletes a user.
PATCH /api/users/:id/toggle-active	Activates or deactivates a user account.

POST /api/users/:id/photo	Uploads a student's photo for face enrollment.
DELETE /api/users/:id/photo	Deletes a student's stored photo.
GET /api/users/:id/embeddings	Retrieves the stored face embeddings of a user.
DELETE /api/users/:id/embeddings	Clears the stored face embeddings of a user.
POST /api/users/clear-db	Clears the user database.

Table 5.2. Student Management API endpoints

Method & Path	Description
GET /api/classes	Retrieves the available classes.
GET /api/classes/upcoming	Retrieves upcoming classes.
GET /api/classes/assigned	Retrieves classes assigned to the authenticated teacher.

GET /api/classes/:id	Retrieves details of a specific class.
POST /api/classes	Creates a new class.
PUT /api/classes/:id	Updates an existing class.
DELETE /api/classes/:id	Deletes a class.
POST /api/classes/:id/students	Adds a student to a class roster.
DELETE /api/classes/:id/students/:studentId	Removes a student from a class roster.

Table 5.3 Class Management API endpoints

Table 5.4. Attendance API endpoints

Method & Path	Description
GET /api/attendance	Retrieves attendance records according to the user's role and access permissions.

GET /api/attendance/stats	Retrieves attendance statistics.
GET /api/attendance/course-summary	Retrieves attendance summaries and session-level course information.
POST /api/attendance/log	Records face-AI based attendance.
POST /api/attendance/batch-session-submit	Submits a completed attendance session using the hybrid AI and manual process.
PATCH /api/attendance/:id/override	Allows an Admin or Teacher to manually override an attendance record.
GET /api/attendance/export/csv	Exports attendance records in CSV format.
GET /api/attendance/export/excel	Exports attendance records in Excel format.

Table 5.4. Attendance API endpoints

Table 5.5. Machine Learning API endpoints

Method & Path	Description
GET /api/ml/health	Checks the health/status of the ML service.
POST /api/ml/start-session	Starts a face-recognition attendance session.
POST /api/ml/stop-session	Stops the active face-recognition session.
POST /api/ml/enroll-face	Enrolls a student's face using the ML service.
POST /api/ml/load-class-db	Loads the selected class's face database for recognition.
POST /api/ml/process-frame	Processes a captured frame for face detection and recognition.
POST /api/ml/clean-class-db	Clears the temporary class face database.
POST /api/ml/clear-all-pkls	Clears all locally stored PKL face databases.

Table 5.5. Machine Learning API endpoints

Method & Path	Description
POST /api/leave	Allows a student to submit a leave request.
GET /api/leave/my	Retrieves the authenticated student's leave requests.
GET /api/leave/teacher	Retrieves leave requests for Teacher/Admin review.
PATCH /api/leave/:id/status	Updates the status of a leave request.

Table 5.6. Leave Management API endpoints

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5.2 ADMINISTRATIVE DATA MANAGEMENT

The screenshots in this segment were captured from the running Smart Attendance Management System and demonstrate the capability of the Admin portal. The portal offers management interfaces for students, teachers, users, classes, and attendance records. Each module shows the relevant data in an organized table and offers the required choices for adding, updating, or managing records. The organizational processes are linked to the corresponding backend API endpoints, and the updated information is displayed in the centralized database. These

interfaces demonstrate how the administrator can manage the core data required for the face-based attendance system

5.2.1 User Directory and Biometrics

The User Directory and Biometrics page (Figure 5.5) offers supervisors with a centralized interface for managing recorded students and their biometric information. The page shows student particulars similar as name, email address, section, roll number, biometric vector status, and account status. Administrators can search and filter users by unit and perform actions similar as adding students, substituting biometric data, editing user information, and managing account records. The interface furthermore signifies whether a student’s biometric vector has been successfully synchronized.

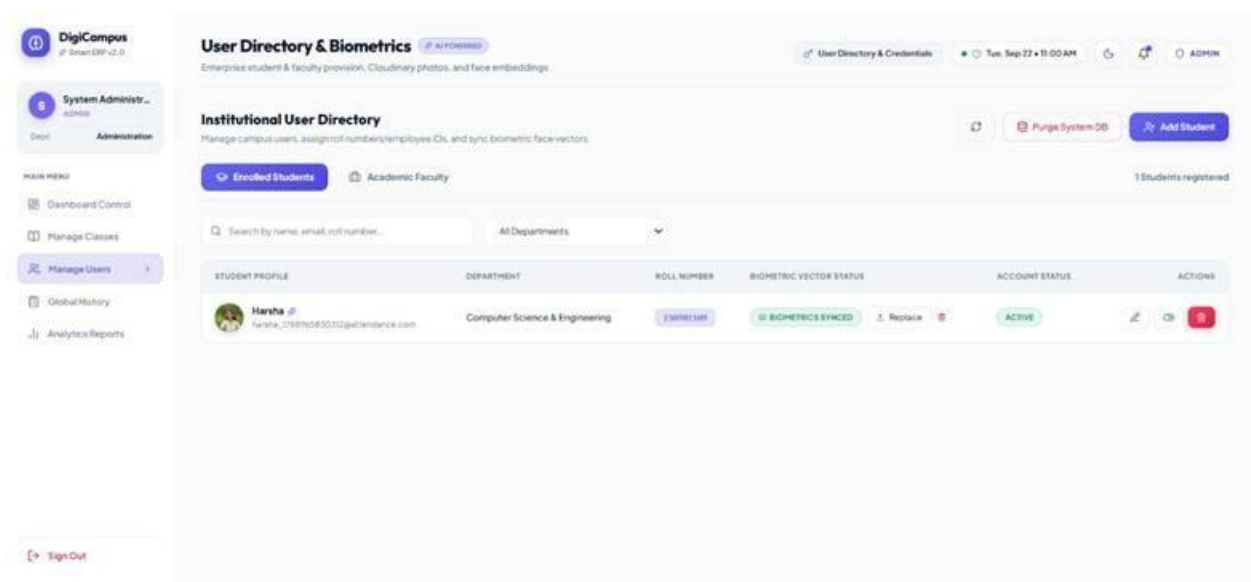


Figure 5.1: Admin User Directory and Biometrics

5.2.2 Admin Master Control and Institutional Hub

The Admin Master Control and Institutional Hub (Figure 5.6) offers an overall summary of the attendance management system. The dashboard shows the

practical status of the Node REST API Engine, MongoDB cluster, and face AI engine, along with essential statistics similar as the total number of recorded users, engaged classes and sections, total presence logs, and typical campus attendance. It moreover presents forthcoming classes and timetable information, permitting supervisors to track the overall operation of the institution's attendance system from a centralized interface.

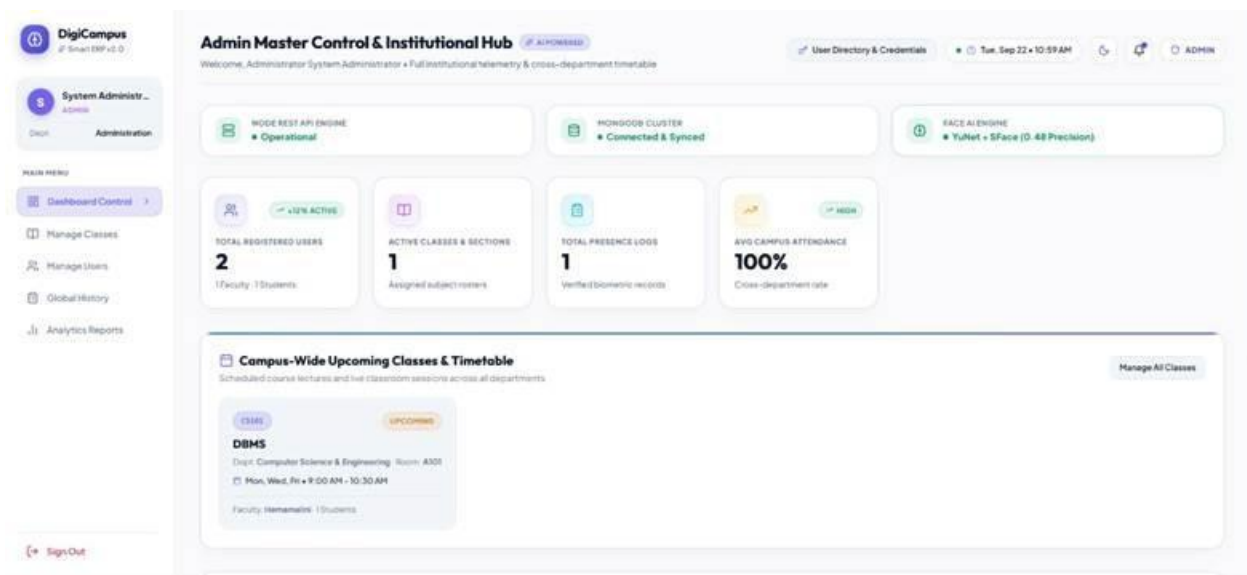


Figure 5.2 Admin Master Control Dashboard

5.3 END-TO-END SYSTEM DEMONSTRATION

The end-to-end presentation illustrates the complete and entire and complete and thorough workflow of the proposed Automated Attendance System across the Teacher and Student portals. The process starts with the teacher performing a live attendance session applying the Live AI Camera and Hybrid Attendance Session (Figure 5.3). The system captures student faces, performs AI-based face detection and recognition, and records the attendance of recognized students. The recorded attendance is then presented through the Analytics and Export Reports module (Figure 5.4), where teachers can view attendance statistics and generate reports.

Students can access their Attendance History (Figure 5.5) and Student Dashboard (Figure 5.6) to observe their attendance percentage, available and absent sessions, and eligibility status. This presentation shows the integration of AI-based face recognition, attendance recording, assessments, reporting, and student

5.3.1 Live AI Camera and Hybrid Attendance Session

The Live AI Camera and Hybrid Session Console (Figure 5.1) offers the instructor with an interface for performing real-time biometric attendance. The instructor can select the required course segment and start an attendance session employing the Start Session choice. The console includes a webcam area for capturing student faces and an AI Frame Detection Log for showing identified students during the session. The interface moreover offers a course roster validation part, permitting the instructor to inspect and adjust student attendance before submitting the last session.

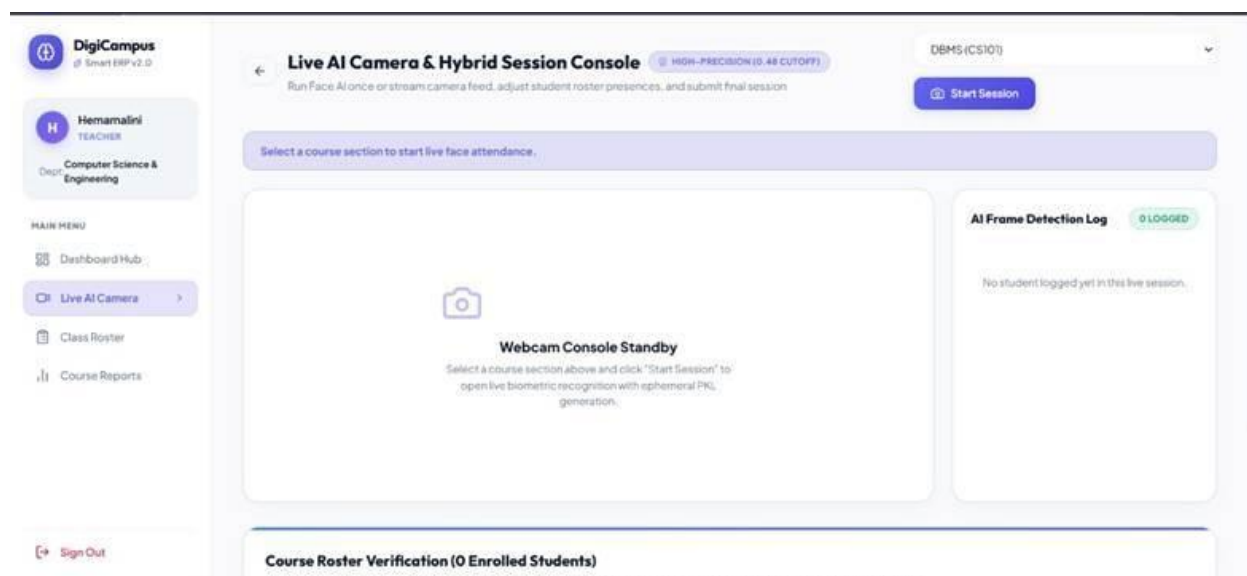


Figure 5.3: Teacher Live AI Camera and Hybrid Session Console

5.3.2 Analytics and Export Reports

The Analytics and Export Reports page (Figure 5.2) offers teachers with a consolidated view of attendance performance. The interface shows the overall attendance rate, total logged events, and functioning course sections, along with visual representations of attendance trends and present-versus-absent statistics. The page furthermore offers choices to export attendance data in CSV and Excel formats, facilitating teachers to download and further examine attendance records.

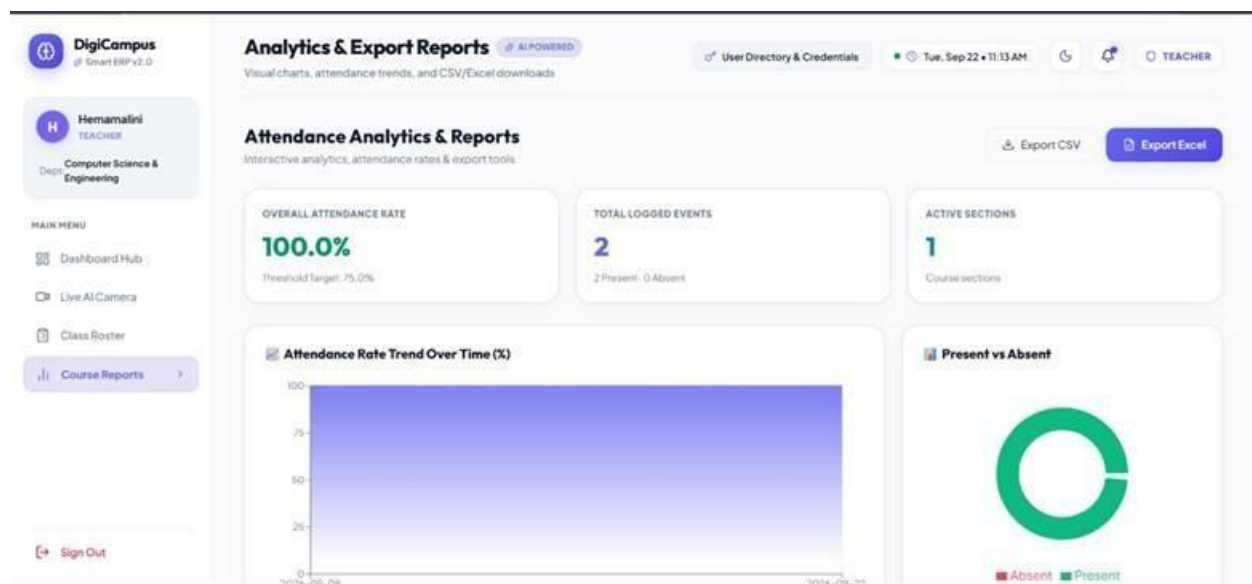


Figure 5.4 Teacher Analytics and Export Reports

5.3.3 Student Attendance History

The Attendance History page (Figure 5.3) offers students with an overview of their planned and finalized classes. The interface shows imminent classes along with particulars similar as the course name, educator, class timetable, room, and registration status. It furthermore offers a course-wise attendance breakdown showing the student's attendance percentage, number of current sessions, absent sessions, and total sessions. This enables students to track their attendance status

across dissimilar courses.

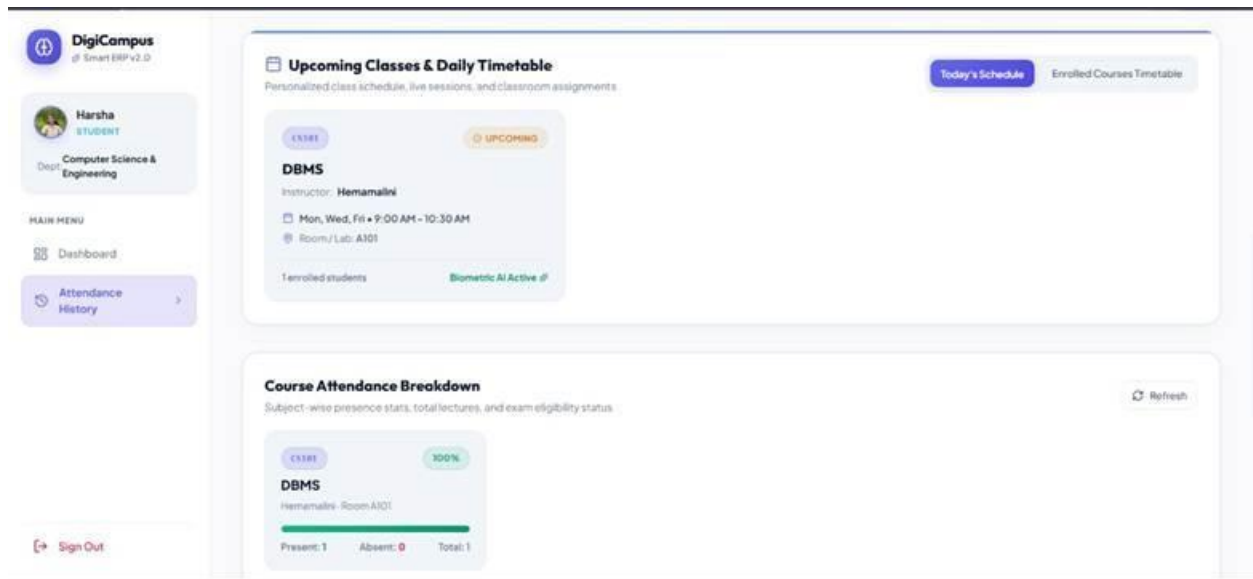


Figure 5.5 Student Attendance History

5.3.4 Student Dashboard and Attendance Portal

The Student Dashboard and Attendance Portal (Figure 5.4) provides students with a centralized view of their attendance information. The dashboard displays the student's overall attendance percentage and cumulative attendance status, along with an indication of examination eligibility based on the configured attendance threshold. Students can view their attendance records and access the On-Duty (OD) application option. The attendance section also provides a summary of total sessions, present sessions, and absent sessions for the selected academic term.

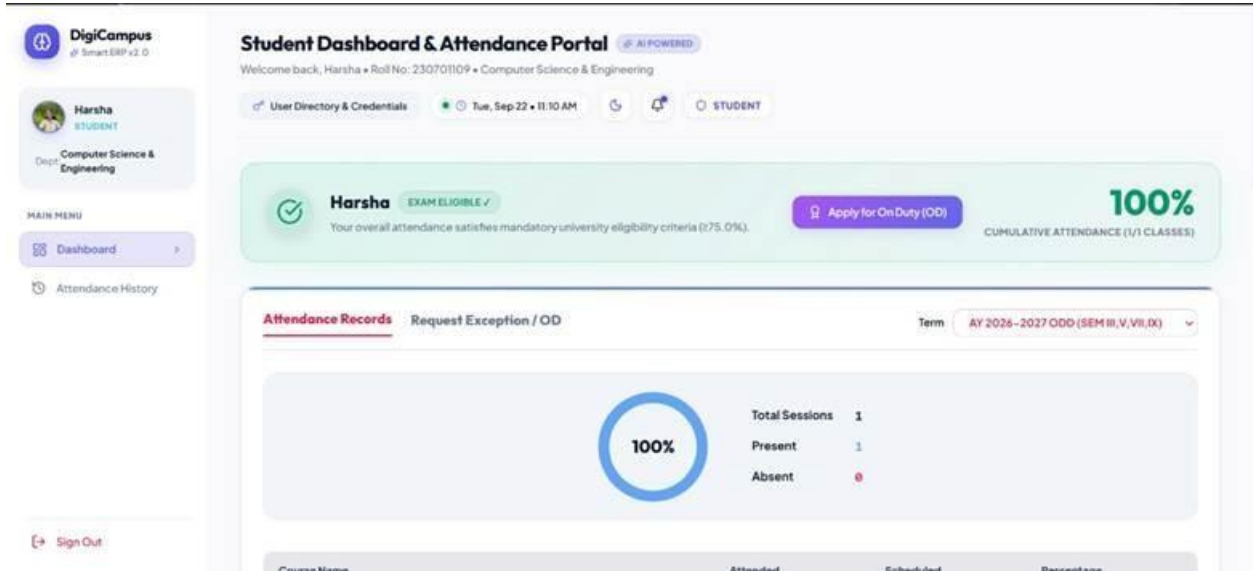


Figure 5.6 Student Dashboard and Attendance Portal

5.4 DISCUSSION

The executed Automated Attendance System shows the integration of AI-based face recognition with a full-stack web application. The Live AI Camera enables teachers to conduct attendance sessions employing face detection and recognition, while the Analytics module offers attendance statistics and reports. The Student portal enables students to observe their attendance history, percentage, and eligibility status. Overall, the system offers a centralized and contactless approach to attendance management, joining AI-based recognition, attendance recording, reporting, and student observation within a separate platform. The end-to-end presentation verifies the functional integration of the important modules, while elaborate and exhaustive accuracy and performance outcomes are discussed individually in the experimental evaluation part.

CHAPTER 6 CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This assignment set out to develop an automated and contactless attendance management system that reduces the nonautomatic effort involved in usual attendance procedures. The executed system achieves this goal through the integration of AI-based face detection and recognition with a full-stack web application. Student biometric information can be enrolled through the Admin or Teacher portal, while the Live AI Camera enables teachers to conduct attendance sessions by detecting and identifying several students. The identified attendance is recorded and made available through centralized dashboards, analyses, and attendance history for teachers and students.

The system merges YuNet for face detection, CLAHE for lighting normalization, Laplacian variance for blur detection, and SFace/ArcFace with ONNX models for face recognition. A Node.js and Express REST API manages authentication, users, classes, attendance, depart, OD, and communication with the Python Flask ML service. The React-based role-specific portals provide separate capability for Admins, Teachers, and Students, while MongoDB Atlas sustains the application's records. The end-to-end presentation in Chapter 5 shows the full and comprehensive workflow from face-based attendance capture to attendance reporting and student tracking.

The undertaking moreover shows multiple pragmatic contributions, comprising contactless attendance, biometric-based student identification, centralized attendance management, attendance assessments, document generation, and role-based entry. The executed system offers a working foundation for automated attendance management while furthermore identifying areas for future

improvement, similar as growing recognition robustness under distinct lighting and face-related conditions, strengthening security and privacy, growing evaluation with bigger real-world datasets, and enhancing the scalability of the deployed system.

6.2 FUTURE WORK.

The future development of the Automated Attendance System can focus on enhancing its accuracy, security, scalability, and real-world usability. The face recognition module can be further improved to improve performance under distinct lighting conditions, face-related angles, camera distances, and fragmentary occlusions. A liveness detection mechanism can moreover be integrated to reduce the possibility of attendance being recorded employing photographs or recordings.

The system can be extended to support bigger organizations and numerous and manifold classrooms concurrently by enhancing the backend architecture and optimizing the face-recognition pipeline. Additional features similar as real-time notifications, advanced attendance analyses, automated warnings for low attendance, and enhanced document generation can furthermore be integrated.

Future versions can further enhance data privacy and security by implementing more robust protection for biometric information, protected communication, and controlled admittance to face embeddings. The system can moreover be assessed utilizing a bigger and more mixed and assorted real-world dataset to measure recognition accuracy, false approval, false rejection, and processing time under dissimilar conditions.

Finally, the application can be extended with mobile support, cloud-based deployment, and automated timetable integration, making the system more proper and right for large-scale structural deployment.

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